

Decimal Ground Truth Accuracy Analysis

How well do models predict pairs when the ground truth is expressed on a finer decimal scale (−2 to +2 in 0.5 steps) vs the 3-class labels they were trained to produce (−1, 0, +1).

Decimal GT Value Distribution

Value	Count	%
-2.0	1	0.0%
-1.5	1	0.0%
-1.0	17	0.3%
-0.5	110	2.0%
+0.0	4588	81.6%
+0.5	354	6.3%
+1.0	419	7.4%
+1.5	46	0.8%
+2.0	89	1.6%

Integer GT Distribution for Half-Value Pairs

The decimal GT is an average of multiple human raters; the integer GT is the final consensus label. This table shows how half-value pairs mapped to the integer GT used for all accuracy measurements.

Decimal GT	n	Integer GT = −1	Integer GT = 0	Integer GT = +1
+0.5	354	6 (1.7%)	25 (7.1%)	323 (91.2%)
−0.5	110	71 (64.5%)	34 (30.9%)	5 (4.5%)

Key asymmetry: Positive half-values were overwhelmingly rounded up to +1 in the integer GT (91.2%), making the `positive_half` mask effectively a set of consensus positive pairs — explaining why models perform reasonably on them. Negative half-values were split: 64.5% became −1 but 30.9% became 0, reflecting genuine annotator disagreement about weak anti-synergies. This label ambiguity is the primary driver of the catastrophically low accuracy on the `negative_half` mask across all models.

Mask Definitions

Mask	Decimal GT values	Expected prediction
neutral	[0.0]	+0
positive_half	[0.5]	+1
positive_clear	[1.0, 1.5, 2.0]	+1
all_positive	[0.5, 1.0, 1.5, 2.0]	+1
negative_half	[-0.5]	-1
negative_clear	[-1.0, -1.5, -2.0]	-1
all_negative	[-0.5, -1.0, -1.5, -2.0]	-1

Mask: neutral

Decimal GT values: [0.0] → expected model output: **+0**

Pairs in mask: **4588**

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	4588	83.7%	98.3%	84.0%	90.6%	33.5%	283 (6%)	3841 (84%)	464 (10%)
gpt54	4588	84.2%	98.2%	84.5%	90.8%	32.7%	392 (9%)	3863 (84%)	333 (7%)
gemini10pro	4588	35.7%	98.2%	35.8%	52.5%	8.4%	419 (9%)	1638 (36%)	2499 (54%)
gemini15flash	4588	83.4%	98.0%	83.5%	90.2%	32.7%	203 (4%)	3828 (83%)	557 (12%)
gpt4omini	4588	69.2%	98.3%	69.4%	81.4%	22.6%	183 (4%)	3173 (69%)	1231 (27%)
gpt4ominift	4588	86.7%	98.2%	86.9%	92.2%	33.3%	139 (3%)	3979 (87%)	470 (10%)
gemini3flash	4588	84.6%	98.2%	84.9%	91.1%	33.9%	370 (8%)	3882 (85%)	336 (7%)

Mask: positive_half

Decimal GT values: [0.5] → expected model output: **+1**

Pairs in mask: **354**

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	354	54.0%	95.3%	56.3%	70.8%	29.7%	13 (4%)	150 (42%)	191 (54%)
gpt54	354	46.0%	93.9%	47.4%	63.0%	27.0%	17 (5%)	174 (49%)	163 (46%)
gemini10pro	354	71.5%	90.5%	70.9%	79.5%	14.5%	17 (5%)	79 (22%)	253 (71%)
gemini15flash	354	28.5%	94.1%	29.4%	44.8%	22.3%	18 (5%)	235 (66%)	101 (29%)
gpt4omini	354	49.7%	92.6%	50.5%	65.3%	31.6%	11 (3%)	167 (47%)	176 (50%)
gpt4ominift	354	33.9%	98.3%	36.5%	53.3%	22.9%	14 (4%)	220 (62%)	120 (34%)
gemini3flash	354	63.6%	94.7%	65.9%	77.7%	33.5%	4 (1%)	125 (35%)	225 (64%)

Mask: positive_clear

Decimal GT values: [1.0, 1.5, 2.0] → expected model output: **+1**

Pairs in mask: **554**

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	554	85.7%	100.0%	85.9%	92.4%	31.7%	9 (2%)	70 (13%)	475 (86%)
gpt54	554	85.2%	100.0%	85.4%	92.1%	31.6%	9 (2%)	73 (13%)	472 (85%)
gemini10pro	554	88.3%	100.0%	88.4%	93.9%	23.5%	18 (3%)	46 (8%)	489 (88%)

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gemini15flash	554	65.5%	100.0%	65.6%	79.3%	26.8%	8 (1%)	183 (33%)	363 (66%)
gpt4omini	554	84.1%	99.8%	84.1%	91.3%	30.4%	7 (1%)	81 (15%)	466 (84%)
gpt4ominift	554	65.2%	100.0%	65.3%	79.0%	26.7%	6 (1%)	187 (34%)	361 (65%)
gemini3flash	554	88.6%	100.0%	88.8%	94.1%	32.6%	9 (2%)	54 (10%)	491 (89%)

Mask: all_positive

Decimal GT values: [0.5, 1.0, 1.5, 2.0] → expected model output: +1

Pairs in mask: 908

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	908	73.3%	98.6%	75.0%	85.2%	33.0%	22 (2%)	220 (24%)	666 (73%)
gpt54	908	69.9%	98.4%	71.3%	82.7%	32.2%	26 (3%)	247 (27%)	635 (70%)
gemini10pro	908	81.7%	96.8%	82.0%	88.8%	15.7%	35 (4%)	125 (14%)	742 (82%)
gemini15flash	908	51.1%	98.7%	52.3%	68.4%	27.7%	26 (3%)	418 (46%)	464 (51%)
gpt4omini	908	70.7%	97.8%	71.7%	82.7%	34.5%	18 (2%)	248 (27%)	642 (71%)
gpt4ominift	908	53.0%	99.6%	54.7%	70.6%	26.6%	20 (2%)	407 (45%)	481 (53%)
gemini3flash	908	78.9%	98.3%	80.4%	88.4%	35.3%	13 (1%)	179 (20%)	716 (79%)

Mask: negative_half

Decimal GT values: [-0.5] → expected model output: -1

Pairs in mask: 110

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	110	26.4%	65.5%	26.8%	38.0%	27.9%	29 (26%)	71 (65%)	10 (9%)
gpt54	110	39.1%	58.1%	35.2%	43.9%	30.6%	43 (39%)	56 (51%)	11 (10%)
gemini10pro	110	25.5%	67.9%	26.8%	38.4%	25.6%	28 (25%)	30 (27%)	52 (47%)
gemini15flash	110	23.6%	61.5%	22.5%	33.0%	27.4%	26 (24%)	62 (56%)	22 (20%)
gpt4omini	110	21.8%	50.0%	16.9%	25.3%	19.4%	24 (22%)	52 (47%)	34 (31%)
gpt4ominift	110	15.5%	52.9%	12.7%	20.5%	21.8%	17 (15%)	77 (70%)	16 (15%)
gemini3flash	110	51.8%	68.4%	54.9%	60.9%	39.2%	57 (52%)	46 (42%)	7 (6%)

Mask: negative_clear

Decimal GT values: [-1.0, -1.5, -2.0] → expected model output: -1

Pairs in mask: 19

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	19	73.7%	100.0%	77.8%	87.5%	29.2%	14 (74%)	4 (21%)	1 (5%)
gpt54	19	63.2%	91.7%	61.1%	73.3%	24.4%	12 (63%)	4 (21%)	3 (16%)
gemini10pro	19	42.1%	100.0%	44.4%	61.5%	27.2%	8 (42%)	2 (11%)	9 (47%)

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gemini15flash	19	47.4%	100.0%	50.0%	66.7%	38.9%	9 (47%)	7 (37%)	3 (16%)
gpt4omini	19	52.6%	100.0%	55.6%	71.4%	23.8%	10 (53%)	3 (16%)	6 (32%)
gpt4ominift	19	47.4%	100.0%	50.0%	66.7%	44.4%	9 (47%)	8 (42%)	2 (11%)
gemini3flash	19	84.2%	93.8%	83.3%	88.2%	29.4%	16 (84%)	1 (5%)	2 (11%)

Mask: all_negative

Decimal GT values: [-0.5, -1.0, -1.5, -2.0] → expected model output: -1

Pairs in mask: 129

Model	n	Accuracy	Prec (target)	Recall (target)	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	129	33.3%	76.7%	37.1%	50.0%	31.3%	43 (33%)	75 (58%)	11 (9%)
gpt54	129	42.6%	65.5%	40.4%	50.0%	31.3%	55 (43%)	60 (47%)	14 (11%)
gemini10pro	129	27.9%	75.0%	30.3%	43.2%	27.5%	36 (28%)	32 (25%)	61 (47%)
gemini15flash	129	27.1%	71.4%	28.1%	40.3%	30.7%	35 (27%)	69 (53%)	25 (19%)
gpt4omini	129	26.4%	64.7%	24.7%	35.8%	22.4%	34 (26%)	55 (43%)	40 (31%)
gpt4ominift	129	20.2%	69.2%	20.2%	31.3%	27.2%	26 (20%)	85 (66%)	18 (14%)
gemini3flash	129	56.6%	74.0%	60.7%	66.7%	39.8%	73 (57%)	47 (36%)	9 (7%)

Half-value vs Clear-value Accuracy Summary

Model	Pos half acc	Pos clear acc	Neg half acc	Neg clear acc
gpt4o	54.0% (n=354)	85.7% (n=554)	26.4% (n=110)	73.7% (n=19)
gpt54	46.0% (n=354)	85.2% (n=554)	39.1% (n=110)	63.2% (n=19)
gemini10pro	71.5% (n=354)	88.3% (n=554)	25.5% (n=110)	42.1% (n=19)
gemini15flash	28.5% (n=354)	65.5% (n=554)	23.6% (n=110)	47.4% (n=19)
gpt4omini	49.7% (n=354)	84.1% (n=554)	21.8% (n=110)	52.6% (n=19)
gpt4ominift	33.9% (n=354)	65.2% (n=554)	15.5% (n=110)	47.4% (n=19)
gemini3flash	63.6% (n=354)	88.6% (n=554)	51.8% (n=110)	84.2% (n=19)

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